

Topology

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based on course by Upendra Kulkarni

- There will be homework which will be credited based on credible completion (not correctness)
- There will be 2 tests/midsem and 1 endsem
- There will be some quizzes

Moore Method: Prof. Moore used to hide all relevant textbooks and provide a list of definitions and results and call on kids. Prof. Kulkarni wants to use them for 1 lecture and tutorials (for some reason).

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1. Upendra Quotes

- “So how many times have you studied Topology? {3} Well, the first time was my fault. Well, this one is not mine.”
- “I have learnt my lesson. I will try not to cause any heartburn anymore.”
- “We will not need to write this unless dealing with some of the ... wrong headed examples. This book (Munkers) abhors in such examples.”
- “In some way, I don’t know why this course exists. Like they say controversial statements to increase TRP.”
- “How many spaces do we care about which don’t arise from metric spaces? Like there are some very important examples but do they warrant a whole course, I think not.”
- “Go and read the definiton of box topology... and throw it in the dustbin”

2. Lecture 1, 6 January 2026

2.1. So what is Topology about?

Topology is not about open spaces, it is about continuous functions. Just like linear algebra is not about vectors and vector spaces but about linear transforms.

Definition: Topology on a Set X

Topology on a set X is a subset of $P(X)$ closed under arbitrary unions and finite intersections.

Remark

Let $\{S_\alpha \mid \alpha \in J\}$ be an indexed family of subsets of X . Then

$$\bigcup_{\alpha \in J} S_\alpha = \{x \in X \mid \exists \alpha \in J \text{ s.t. } x \in S_\alpha\}$$

$$\bigcap_{\alpha \in J} S_\alpha = \{x \in X \mid \forall \alpha \in J, x \in S_\alpha\}$$

This implies every topology on X contains X and $\emptyset$.

We normally denote a topological space as (X, τ) but we ignore the τ for most cases other than some counter-examples.¹

We will study more about geometric topology and glueing etc.

Example

- For any X , discrete topology refers to $\tau = P(X)$ while indiscrete topology refers to $\tau = \{\emptyset, P(X)\}$.
- Finite Complement Topology²
- Countable Complement Topology
- The usual topologies of $\mathbb{R}, \mathbb{R}^n$, any metric spaces.

Definition: Basis for a Topology

The basis for a topology on X is a subset $\mathcal{B}$ of $P(X)$ such that

$$\bigcup_{v \in \mathcal{B}} v = X$$

Also, given $x \in B_1 \cap B_2, B_i \in \mathcal{B}$:

¹“We will not need to write this unless dealing with some of the ... wrong headed examples. This book (Munkers) abhors in such examples.”

²Will come in as the Zariski Topology for the affine line, words I don't know the meaning of

$$\exists B_3 \in \mathcal{B} \text{ s.t. } x \in B_3 \subset B_1 \cap B_2$$

Given an abstract basis $\mathcal{B}$ in $P(X)$, the topology generated by $\mathcal{B}$ consists of those sets U such that $\forall x \in U, \exists B \in \mathcal{B} \text{ s.t. } x \in B \subseteq U$.

Remark 2.1. This gives a topology. Also, note: each basis element is open.

Theorem 2.2. U is open in this topology $\iff U$ is union of basis elements

Proof.

TODO. ?

■

Definition: Subbasis

An abstract subbasis is any subset $\mathcal{S}$ of $P(X)$ such that

$$\bigcup_{v \in \mathcal{S}} v = X$$

then the associated basis is set of all finite intersections of elements of $\mathcal{S}$.

Suppose I am separately given τ , a topology on X and $\mathcal{B}$ is an abstract basis in $P(X)$. We say $\mathcal{B}$ generates $\tau \iff$ if each $B \in \mathcal{B}$ is in τ and for any $U \in \tau$ is union of elements in $\mathcal{B}$.

Definition

If τ_1, τ_2 are two topologies on X , we say τ_2 is finer than τ_1 and τ_1 is coarser than τ_2 if $\tau_1 \subseteq \tau_2$.

2.2. (X, d) metric space

Take $\mathcal{B}$ = open balls, $B_{\varepsilon(a)} = \{x \in X \mid d(x, a) < \varepsilon\}$. This is a basis by triangle inequality.

Example

- $\mathbb{R}$ has basis (a, b)
- $\mathbb{R}^2$ has basis balls (l_2) as well as diamonds (l_1) (we have already seen that all norms are equivalent on finite dimensional $\mathbb{R}^n$)

3. Lecture 2, January 7 2026

TODO.

4. Lecture 3, January 8 2026

4.1. Continuity

Definition 4.1. $X \xrightarrow{f} Y$ is continuous

$\Leftrightarrow f^{-1}(\text{open in } Y) \text{ is open in } X$

$\Leftrightarrow f^{-1}(\text{closed in } Y) \text{ is closed in } X$

$\Leftrightarrow f(\overline{A}) \subset \overline{f(A)}$

$\Leftrightarrow f$ is continuous at each $x \in X$

Remark 4.2. To determine if f is cont, it is enough to check $f^{-1}(\text{basis elements})$ is open.

Corollary 4.3. $X \xrightarrow{f} Y$ is continuous $\Rightarrow X_{\text{finer}} \xrightarrow{f} Y$ is continuous

Example

Following are continuous:

- Any X , $X \xrightarrow{\text{id}} X$
- $X \xrightarrow{f} Y \xrightarrow{g} Z$ where f, g is continuous then $g \circ f$ is continuous.
- any $X \xrightarrow{\text{constant}} \text{any } Y$
- Inclusion of any subspace
- restriction of domain
- extend codomain or restrict codomain if possible
- local formulation of continuity. If $\{V_\alpha\}_{\alpha \in I}$ is an open cover of X then $X \xrightarrow{f} Y$ is continuous $\Leftrightarrow f|_{V_\alpha} : V_\alpha \rightarrow Y$ is continuous $\forall \alpha$
- Pasting Lemma for finitely many closed sets

4.2. Product Topology

Let's say we somehow have

$$\mathbb{R} \times \mathbb{R} \xrightarrow{+, \cdot} \mathbb{R}$$

$$\mathbb{R} \xrightarrow{-} \mathbb{R}$$

$$\mathbb{R} \setminus \{0\} \xrightarrow{\frac{1}{x}} \mathbb{R}$$

are continuous. This gives as many more. Say we want to show x^2 is continuous

$$\begin{aligned}\mathbb{R} &\xrightarrow{\Delta} \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R} \\ x &\mapsto (x, x) \\ (x, y) &\mapsto xy\end{aligned}$$

Basically, we want that if $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are continuous then $f + g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous. We can do that via the continuity of $f \times g : \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$.

Thus us what we want our product topology to achieve.

Claim 4.4. If $X \xrightarrow{f_i} Y_i$ is continuous then $X \xrightarrow{f_1 \times f_2} Y_1 \times Y_2$ is continuous.

Let's try using $V_1 \times V_2$ where V_i is open in Y_i should be open in $Y_1 \times Y_2$.

$$(V_1 \times V_2) \cap (W_1 \times W_2) = (V_1 \cap W_1) \times (V_2 \cap W_2)$$

but

$$(V_1 \times V_2) \cup (W_1 \times W_2) \neq (V_1 \cup W_1) \times (V_2 \cup W_2)$$

To get the latter to be true, we define it as

Definition

$Y_1 \times Y_2$ becomes a topological space by treating open $\times$ open is a basis.

Furthermore, observe that $Y_1 \times Y_2 \xrightarrow{p_i} Y_i$ being the projection maps are continuous as:

$$\begin{aligned}p_1^{-1}(V_1) &= V_1 \times Y_2 \\ p_2^{-1}(V_2) &= Y_1 \times V_2 \\ p_1^{-1}(V_1) \cap p_2^{-1}(V_2) &= V_1 \times V_2\end{aligned}$$

This implies that the product topology is the coarsest topology on $Y_1 \times Y_2$ that makes projections continuous.

Claim 4.5. $(Y_1 \times Y_2, p_1, p_2)$ is the universal topological space with continuous maps to Y_1 and Y_2 .

This means, given continuous $Z \xrightarrow{f_i} Y_i$ then $\exists! f : Z \rightarrow Y_1 \times Y_2$ such that

$$f_i = p_i \circ f$$

Proof.

TODO.

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Remark 4.6. This leads to the category of topological spaces and continuous map.

4.3. Isomorphism of Topological Spaces

Notice, Isomorphism is just renaming while preserving structure.

Bad Hopeful Definition 4.7. A bijective continuous map

Why is this bad? Consider (X, τ) then

$$(X, \tau) \xrightarrow{\text{id}} (X, \tau')$$

is continuous and bijective if τ is finer. This has by no means preserved structure!

Definition: Isomorphism

An isomorphism is a map

$$\begin{aligned} X &\xrightarrow{f} Y \text{ s.t.} \\ \exists X &\xleftarrow{g} Y \text{ with} \\ fg &= \text{id}_Y, gf = \text{id}_X \end{aligned}$$

(Such g_1 if it exists is unique: $g_2 = (g_1 f)g_2 = g_1$)

Definition 4.8. A continuous map $f : X \rightarrow Y$ is called a homeomorphism if exists a continuous map $g : Y \rightarrow X$ such that $fg = \text{id}_Y, gf = \text{id}_X$